

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1003	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	22/08/2026	Duration:	60 minutes

Code of Conduct

I K Madan Mohan Reddy (192411206) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Byreddy _____

Annexure A – Test Case Design Assignment

“Online Loan Eligibility Prediction and Approval Management Platform”

Your test case design should include:

1. Identify the Functional and Non-Functional Requirements to Be Tested

Answer

The Online Loan Eligibility Prediction and Approval Management Platform should be tested against both functional and non-functional requirements. Functional testing verifies whether the system performs all required loan processing operations correctly, while non-functional testing checks the system’s security, accuracy, performance, reliability, and usability. Functional Requirements to Be Tested

- User registration and login.
- Secure user authentication.
- Collection of applicant personal and financial details.
- Verification of income, employment, and credit information.
- Calculation and display of loan eligibility.
- AI-based prediction of loan approval eligibility.

- Loan application submission and status tracking.
 - Admin review and approval or rejection of applications.
 - Generation of loan eligibility reports.
 - Notification of loan approval or rejection.
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2. Identify the Test Scenarios for the Assigned Problem Statement

Answer

Test scenarios describe the major situations that need to be tested in the Online Loan Eligibility Prediction and Approval Management Platform. They cover both normal system operations and possible errors. The testing ensures that user registration, loan application, eligibility prediction, approval processing, data management, and system performance work correctly.

Major Test Scenarios

- Verify login with valid and invalid credentials.
 - Verify that a new user can register successfully.
 - Verify that applicant personal and financial details are entered correctly.
 - Verify that invalid or incomplete applicant information is rejected.
 - Verify correct loan eligibility calculation for different applicant profiles.
 - Verify that loan eligibility results are displayed correctly.
 - Verify AI-based loan approval prediction.
 - Verify prediction when incomplete applicant data is provided.
 - Verify loan application submission and status tracking..
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3. Prepare Detailed Test Cases Covering Normal, Boundary, Invalid, and Exceptional Conditions

Answer

The test cases are prepared to cover different conditions that may occur during the operation of the Online Loan Eligibility Prediction and Approval Management Platform. Normal test cases verify that the system works correctly with valid applicant and loan details. Boundary test cases check values at the minimum and maximum acceptable limits. Invalid test cases use incorrect or unexpected inputs. Exceptional test cases verify how the system behaves during failures such as database failure, network interruption, or service unavailability.

Test Conditions

Normal: Valid applicant, income, credit score, and loan details are provided.

Boundary: Minimum and maximum eligible income, credit score, and loan amounts are tested.

Invalid: Negative values, text values, missing details, and incorrect credentials are tested.

Exceptional: Database failure, network interruption, and prediction service failure are tested.

4. Include Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status

Answer

The following test cases provide functional coverage for the major modules of the AI-Based Air Quality Monitoring and Prediction System. The Actual Result and Pass/Fail Status can be filled during actual execution of the application.

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-01	User Registration	Open registration → Enter details → Submit	Valid name, email, password	Account should be created	To be recorded	Pass/Fail
TC-02	Valid Login	Enter valid credentials → Click Login	Correct email/password	User should enter dashboard	To be recorded	Pass/Fail
TC-03	Invalid Login	Enter incorrect password → Login	Wrong password	Error message should be displayed	To be recorded	Pass/Fail
TC-04	Sensor Data Collection	Send sensor readings to server	PM2.5=35, PM10=60	Data should be received and stored	To be recorded	Pass/Fail
TC-05	Invalid Sensor Data	Send negative pollution value	PM2.5=-10	System should reject invalid data	To be recorded	Pass/Fail

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-06	AQI Monitoring	Open dashboard	Valid sensor readings	Current AQI should be displayed	To be recorded	Pass/Fail
TC-07	AQI Prediction	Enter historical/current data → Run prediction	Valid environmental data	Future AQI should be predicted	To be recorded	Pass/Fail
TC-08	Missing Prediction Data	Submit incomplete data	Missing PM2.5 value	Appropriate validation message should appear	To be recorded	Pass/Fail
TC-09	Pollution Alert	Provide AQI above configured limit	High AQI value	Pollution alert should be generated	To be recorded	Pass/Fail
TC-10	Safe AQI	Provide AQI within safe range	Normal AQI value	No high-pollution alert should be generated	To be recorded	Pass/Fail
TC-11	Historical Data	Select previous date → Search	Valid date	Historical AQI data should be displayed	To be recorded	Pass/Fail
TC-12	Report Generation	Select period → Generate Report	Monthly data	Report should be generated successfully	To be recorded	Pass/Fail
TC-13	Sensor Failure	Disconnect sensor → Monitor system	No sensor connection	System should show sensor error/status	To be recorded	Pass/Fail
TC-14	Database Failure	Stop database → Request data	Database unavailable	System should show suitable error message	To be recorded	Pass/Fail

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-15	Multiple Users	Access system using multiple sessions	Multiple user requests	System should remain responsive	To be recorded	Pass/Fail

5. Apply Appropriate Test Design Techniques

Answer

Different test design techniques are applied to improve the effectiveness and coverage of testing. These techniques help identify errors using a smaller but meaningful set of test cases.

5.1 Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups. For example, AQI values can be divided into safe, moderate, unhealthy, and hazardous categories. Instead of testing every possible AQI value, one or more representative values from each category can be tested.

Example:

- Valid AQI → 50
- Moderate AQI → 100
- High AQI → 200
- Invalid AQI → -10

5.2 Boundary Value Analysis

Boundary Value Analysis checks values at the edges of an acceptable range. It is useful for testing AQI thresholds and sensor input limits.

For example, if an alert is generated when AQI reaches a configured limit, the values immediately below, equal to, and above that limit should be tested.

Example:

- Limit - 1 → No alert
- Limit → Alert according to configured rule
- Limit + 1 → Alert

This helps identify errors related to incorrect comparison conditions.

5.3 Decision Table Testing

Decision Table Testing is useful for testing pollution alerts because multiple conditions may determine whether an alert should be generated.

AQI Condition	Sensor Available	User Alert Enabled	Expected Result
Safe	Yes	Yes	No high-pollution alert
High	Yes	Yes	Send alert
High	Yes	No	Do not send user alert
High	No	Yes	Display sensor/data error

This technique ensures that different combinations of conditions are tested.

5.4 State Transition Testing

State Transition Testing can be used to verify the different states of the monitoring system.

Example:

Sensor Connected



Data Received



AQI Calculated



AQI Monitored



High Pollution Detected



Alert Generated

If the sensor becomes disconnected, the system should move to a Sensor Error/Offline state and notify the administrator when appropriate. This ensures that the system handles changes in operating conditions correctly.

6. Ensure Adequate Requirement and Functional Coverage

Answer

Requirement coverage ensures that every important system requirement is tested by at least one test case. The test cases cover user authentication, sensor data collection, AQI calculation, AI prediction, alerts, historical data, reporting, and system failure conditions. Non-functional requirements such as performance, security, reliability, usability, and availability should also be tested separately. Boundary and invalid inputs are included to ensure that the application does not produce incorrect results. Exceptional conditions such as sensor, database, and network failures are also considered. The test results should be recorded after executing the application. Any failed test case should be investigated and corrected before deployment. Regression testing should then be performed to ensure that the correction has not affected existing functions. This approach provides better confidence that the system meets its requirements.

Requirement Coverage Summary

Requirement	Covered By
User Registration	TC-01
User Authentication	TC-02, TC-03
Sensor Data Collection	TC-04, TC-05
AQI Monitoring	TC-06
AI Prediction	TC-07, TC-08
Pollution Alerts	TC-09, TC-10

Requirement	Covered By
Historical Data	TC-11
Report Generation	TC-12
Sensor Reliability	TC-13
Database Reliability	TC-14
Performance/Scalability	TC-15
Boundary Testing	AQI boundary cases
Invalid Input Testing	TC-03, TC-05, TC-08
Exceptional Testing	TC-13, TC-14

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect

Application of Test Design Techniques(15 %)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25